Analysis of AI virtual influencer marketing strategy based on AISAS model——Taking Liu Yexi as an example

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Abstract

This paper analyzes the marketing strategies of AI virtual influencers using the AISAS model, focusing on the Chinese virtual influencer Liu Yexi. It discusses Liu Yexi's effectiveness in engaging consumers through promotion on social media platforms and explores the consequent consumer behaviors in terms of attention, interest, search, action, and share (AISAS). Through a survey of 306 participants familiar with Liu Yexi and with analysis of structural equation modeling (SEM), this study assesses the impact of virtual influencer promotions and their ability to transition consumers' attention, interest and search into action and share. The findings suggest that promotional efforts of virtual influencers significantly raise consumers' attention, interest and search willingness. And the search willingness can in turn enhance purchasing action which can finally induce consumers to share more on social media.

CCS Concepts

• **Applied computing** → Operations research; Marketing.

Keywords

AI virtual influencer, Social media marketing, AISAS model, Structural equation modeling, LiuYexi

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1 INTRODUCTION

The rapid advancement of artificial intelligence has catalyzed significant innovations in digital marketing, notably through the evolution of digital virtual human technology. This has given rise to "virtual influencers," who are now pivotal in social media marketing. Virtual influencers are fictional entities operated by third parties (freelance creators, digital agencies, or brands)[1]. Currently, millions of consumers interact with virtual influencers on social media platforms; they have nearly three times the engagement rate on Instagram with virtual influencers compared to human influencers[2]. Both luxury and mass brands are willing to collaborate with virtual influencers for related marketing[3].

Undoubtedly, the virtual influencers generated by computer science are rapidly influencing enterprise management, especially in the field of marketing. At present, research has shown that virtual influencers have some advantages that differ from traditional influencers: they can reduce operating costs, be better controlled by brands to minimize the risk of "human errors", give brands a better opportunity to engage in more effective native advertising[4], and help brand owners keep up with the trends of the metaverse to carry out marketing activities[5]. On the other hand, a series of experiments have been conducted, indicating that virtual influencers with different attributes will bring different user reactions to social media platforms, and the language type they use[6] and the product type they endorse[7] will affect their marketing effectiveness. However, as an emerging marketing tool, what kind of marketing effects have the existing virtual influencers exactly brought, and how to further improve them in the future are still urgent issues that need to be solved.

Promotion is a very important part of marketing[8]. Just as products require promotion to convey information—the catalyst of all consumer behavior—virtual influencers also need to promote themselves, with no exception for Liu Yexi. She is one of the most widely followed virtual influencers in China at present, and she has been widely recognized as a beauty expert integrating Chinese style and Cyberpunk technology since her debut on the Douyin platform in 2021, receiving more than three million likes of her first video. With the design concept of multi element IP collision, marketing strategy to cater to the market, and excellent eye-catching video content, Liu Yexi has become a well-known influencer on Chinese social media. Therefore, it is meaningful to focus on Liu Yexi'promotion to study the marketing effectiveness and improvement measures of Chinese virtual influencer.

This paper analyzes a large amount of literature, and based on a clear understanding of the development and characteristics of virtual internet celebrities, focuses on the virtual influencer, Liu Yexi, in China. Through 306 survey results, this analysis reflects how Liu Yexi's promotion on social media platforms affects the attention, interest, search, action, and share (AISAS) aspects of consumer responses. Based on this, the study evaluates its marketing strategies and offers suggestions for enhancement.

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Figure 1: AISAS theoretical model.

2 RESEARCH FRAMEWORK AND HYPOTHESES

2.1 AISAS theoretical model

The AISAS model is a theoretical model in the field of consumer behavior that evolved from the traditional AIDMA model for online consumers. In traditional marketing, the AIDMA model assumes that consumers go through five stages from perceiving information to the final transaction: attention, interest, desire, memory, and action. However, in the context of digital marketing in the Internet era, the AIDMA model has been unable to fully explain the new behavior of consumers. Therefore, in 2004, Japan Telecom Corporation proposed the AISAS theory as an extension of the AIDMA model, introducing two additional behaviors typical of the Internet era—search and share. This update expanded the traditional five-stage model of consumer purchase decisions—attention, interest, desire, memory, and action—to include these new stages, creating a model now comprising attention, interest, search, action, and share. The framework of AISAS theoretical model shows in Figure 1 below.

2.2 Research framework

Building on literature that develops a systematic AISAS model linked with promotional factors[9], this study, incorporating an analysis of Liu Yexi, proposes a framework that aims to illustrate consumer responses to promotions by virtual influencers, as shown in Figure 2. This study includes seven hypotheses.

H1: Promotion of virtual influencers increases consumers' attention.

H2: Promotion of virtual influencers enhanced consumers' interest.

H3: Promotion of virtual influencers boosts consumers' information search willingness.

H4: Increased consumers' attention to virtual influencers' promotions enhances the likelihood of taking purchase actions.

H5: Elevated consumers' interest in virtual influencers' promotions boosts the likelihood of taking purchase actions.

H6: Greater consumers' information search willingness to virtual influencers' promotions increases likelihood of taking purchase actions.

H7: Purchase actions for products recommended by virtual influencers heighten the likelihood of consumers sharing related information on social media.

3 METHOD

To test the hypotheses, this study employs descriptive analysis supported by Structural Equation Modeling (SEM). This study uses primary data and employs descriptive analysis as its methodological approach. Sampling is conducted through purposive judgment, selecting participants specifically suited to the research objectives.

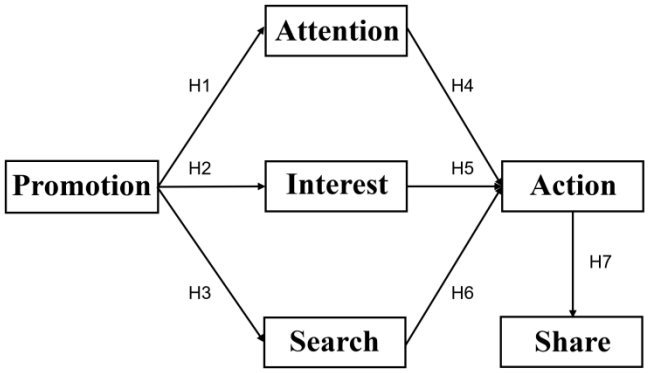

Figure 2: Research framework.

3.1 Participants

The questions in the questionnaire consisted of closed questions consisting of 29 questions about familiarity about Liu Yexi, promotion, attention, interest, search, action, share and demographics. This study selected participants from the Credamo platform, which is one of the largest online survey providers in China. Of the 312 respondents, 306 participants passed the test of whether they knew Liu Yexi and had experiences buying products recommended by her, as indicated by an item on a two-option rating scale (1 = Yes, 0 = No) "I knew the virtual influencer Liu Yexi, and I used to buy products recommended by Liu Yexi", so data from 306 participants were used for analysis. Of the respondents, 67.6% were female, 35.2% were below 25 years of age, 18.3% were between 25 and 30 years of age, and 37.5% were between 31 and 40 years of age. In terms of education, 83.3% of participants had graduated from junior colleges and universities, 5.5% were below junior college, and 11.1% were at graduate level or above. For annual income, 48.3% made RMB 100,000 - 250,000, and 25.1% made RMB 250,001 - 400,000.

3.2 Variables and Indicators

There are 6 latent variables used in this study, namely Promotion (PR), Attention (AT), Interest (IT), Search (SE), Action (AC), and Share (SH). Promotion variables are independent variables, while the variables of attention, interest, search, action and share are dependent variables. Except for the variable of Search, which is influenced by three indicators, every variable is influenced by four indicators[9,10]. Details on the variables, dimensions, indicators, Cronbach's Alpha value, and the average scores of each indicator, assessed using a 7-point Likert scale, are presented in Table 1.

Table 1: Model Descriptions.

Variable	Dimension	Indicator	Code	Average score	Cronbach's α
Promotion	Use social media for promoting the AI virtual influencer	1. Self-Creation and Publication	PR1	5.26	0.699
		2. UGC (User-Generated Content)	PR2	5.20	
		3. Public Relation	PR3	5.91	
		4. Publicity	PR4	5.79	
Attention	Pay attention to the AI virtual influencer	1. Intensity of Attention	AT1	5.56	0.735
		2. Preference of Platform	AT2	5.64	
		3. Impression of Exposure	AT3	5.60	
		4. Coherence of content	AT4	5.63	
Interest	Interested in the AI virtual influencer's content	1. Interest in Content	IN1	5.70	0.800
		2. Interest in Marketing Activities	IN2	5.44	
		3. Preference for Type of Activities	IN3	5.61	
		4. Interest in Promoted products	IN4	5.68	
Search	Search information about the AI virtual influencer's content	1.Further Search Frequency	SE1	4.92	0.736
		2. Value of Search	SE2	5.23	
		3. Accessibility of Search information	SE3	5.57	
Action	Take purchase actions	1. Purchase after seeing the content	AC1	5.48	0.759
		2. Purchase after being interested in the content	AC2	5.82	
		3. Purchase after searching for additional information	AC3	5.78	
		4. Purchasing experience	AC4	5.88	
Share	Share information through social media	1. Give Testimony	SH1	5.65	0.763
		2. Initiative share on social media	SH2	5.27	
		3. Share with material reward on social media	SH3	5.71	
		4. Share with other people you know	SH4	5.58	

Table 2: KMO and Bartlett's test.

KMO value		0.953
Bartlett sphericity test	Approx. Chi-Square	3398.300
	df	253
	P value	0.000

4 RESULT

4.1 Evaluation of measurement model indicators

This study used SPSS 26.0 to test the questionnaire's reliability and validity. The reliability of the questionnaire is often judged by Cronbach's Alpha value in the process of questionnaire analysis. When Cronbach's Alpha values are greater than 0.6, the questionnaire can be proved to have considerable reliability. It can be concluded from Table 1 that Cronbach's Alpha values of all variables are greater than 0.6, indicating that the questionnaire in this paper has considerable reliability and can be analyzed in the next step.

The validity of the questionnaire is often analyzed by Kaiser-Meyer-OIkin measurement and Bartlett's test. When the KMO value is greater than 0.8, it shows that the reliability of the questionnaire is very good, and the original variables are more suitable for factor analysis. For Bartlett's spherical test, if the SPSS test results show Sig.<0.001, it indicates that it meets the standard, the data is spherical distribution, and each variable is independent of each other to a certain extent. According to Table 2, the KMO value is 0.953, greater than 0.8, the approximate Chi-Square value of Bartlett sphericity test is 3398.3, and the significance probability (P value) is less than 0.001, which indicates that the questionnaire in this paper has good validity.

4.2 SEM model test and hypotheses testing

The description of the Table 3 is intended to measure the suitability of the model used, which includes Minimum Discrepancy (CMIN), Degree of Freedom (DF), CMIN/DF, Root Mean Square Residual (RMR), Root Mean Square The error of Approximation (RMSEA), Goodness of Fit (GFI), Comparatives Fit Index (CFI), Incremental Fit Index (IFI), and Tucker-Lewis (TLI). Based on the values of the Table 3, most of the values meet the suitability requirements of the model,

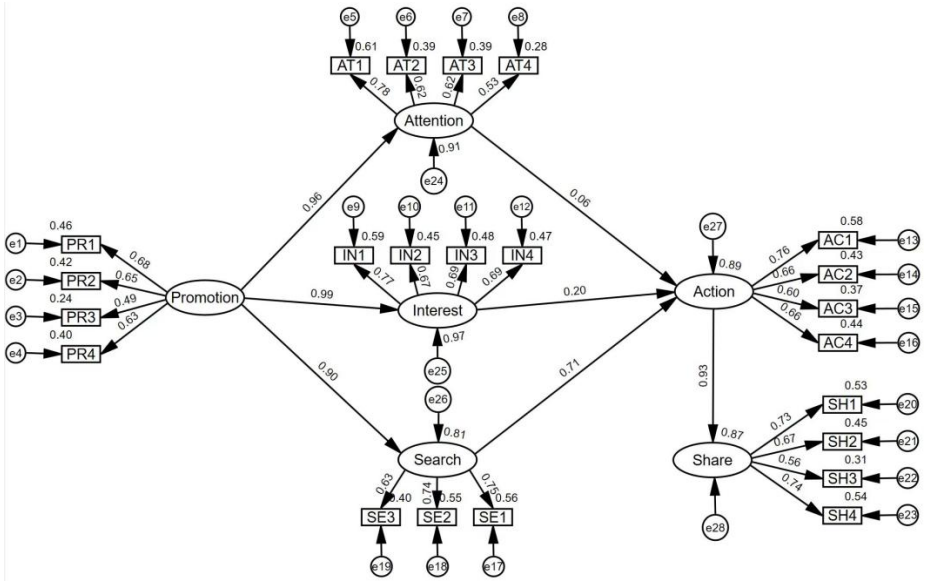

Figure 3: The standardized path analysis of SEM.

Table 3: The model compatibility test.

Indicator	Cut off value	Result
CMIN		532.131
DF		223
CMIN/DF	≤3	2.386
RMR	≤0.05 atau ≤0.1	0.061
RMSEA	≤0.05 atau ≤0.08	0.067
GFI	≥0.9	0.850
CFI	≥0.9	0.905
IFI	≥0.9	0.906
TLI	≥0.9	0.892

only 2 values marginally fit the cut off value, so the conclusion is that in general, the model has a good level of compatibility.

Based on the good compatibility, this study used AMOS 28.0 to conduct SEM model analysis and hypotheses test. By using AMOS to analyze the structural equation modeling in this paper, the path coefficients showed the interrelationships and degree of influence between latent variables, as well as between latent variables and their indicators. The hypothesis testing results is shown in Table 4 and the standardized path analysis diagram of the structural equation modeling is shown in Figure 3. Table 4 shows the significance level (β), critical ratio (C.R.), significance probability (P value) of every path, and the conclusions of all hypotheses. Figure 3 shows the standardized path coefficient of the model.

The impact of promotion of virtual influencers: according to the results of Table 4 and Figure 3, promotion of virtual influencers had significant and positive impact on comsumers' attention (β =0.956, C.R.>1.96, P<0.001), so H1 passed the test. And, promotion of virtual influencers had significant and positive impact on comsumers' interest (β =0.987, C.R.>1.96, P<0.001), so H2 passed the test. In addition, the positive impact of promotion on consumers' information search willingness (β =0.900, C.R.>1.96, P<0.001) was verified, so H3 was supported.

The impact of comsumers' attention on purchase action: according to the results of Table 4 and Figure 3, comsumers' attention didn't have significant impact on consumers' decision to take purchase actions (β =0.062, C.R.=0.228, P=0.820), so H4 was not supported.

The impact of comsumers' interest on purchase action: according to the results of Table 4 and Figure 3, comsumers' interest didn't

Table 4: Hypothesis testing results.

Hypotheses	Path	β	C.R.	P value	Conclusion
H1	Promotion → Attention	0.956	11.807	***	Support
H2	Promotion → Interest	0.987	11.743	***	Support
H3	Promotion → Search	0.900	10.845	***	Support
H4	Attention → Action	0.062	0.228	0.820	Not Support
H5	Interest → Action	0.195	0.575	0.566	Not Support
H6	Search → Action	0.709	3.051	0.002	Support
H7	Action → Share	0.932	11.888	***	Support

have significant impact on consumers' decision to take purchase actions (β =0.195, C.R.=0.575, P=0.566). So H5 didn't pass the test.

The impact of comsumers' information search willingness on purchase action: according to the results of Table 4 and Figure 3, the positive impact of comsumers' information search willingness on consumers' decision to take purchase actions (β =0.709, C.R.>1.96, P<0.005) was verified. Therefore, H6 was supported.

The impact of purchase action on consumers' desire to share: according to the results of Table 4 and Figure 3, purchase actions of products the virtual influencer recommended influenced the desire of consumers to share related information through social media positively and significantly (β =0.932, C.R.>1.96, P<0.001). Therefore, H7 passed the test.

5 DISCUSSION

This study focused on the Chinese AI virtual influencer Liu Yexi to discuss the effectiveness of AI virtual influencers' promotion and their current marketing situation. Based on the AISAS theory model, the five variables were chosen to measure consumers' responses at different stages: attention, interest, search, action and share. According to the results above, there are several findings of this study.

First, Liu Yexi's promotion was proven to be effective. The scores related to promotion in Table 1 were all high means this AI virtual influencer did well in self-creation and publication, attracted many social media users and medium to create content related to her, and she also established positive social connections and a good social media image. Additionally, the results of analysis of SEM showed Liu's promotion strongly raised consumers' attention and interest to her, and also raised consumers' information search willingness. So no matter whether these impact would eventually increase consumers' purchase intention of the products' Liu recommended, the results at least shows that Liu Yexi's self-promotion could bring more popularity to her own, which also helps to explain why she can become a big influencer in such a high speed and in such a short time.

Second, the results emphasize the importance of consumers' information search willingness related to virtual influencers. It's obvious that Liu Yexi has done well in attracting attentuon, arousing interest and stimulating people's willingness to search for information. However, only by strengthening consumers' willingness to search for information can consumers' purchasing behavior be effectively promoted, which cannot be achieved only by arousing attention and interest.

Third, purchase actions of products recommended by virtual influencers can raise the desire of consumers to share related information through social media. The Table 1 shows that simply attracting attention is less likely to induce consumers to purchase. Participants thought they could have good purchasing experience by buying products recommended by Liu Yexi. Then, these action's indicators made positive influence on consumers' desire of sharing information on social media. The significance of this result is that it shows the mutual influence between the popularity of virtual influencer and purchasing action, because the promotion of virtual influencer enhances consumers' purchasing action by improving consumers' willingness to search for information, and this action can prompt consumers to share information online; in other words, it promotes consumer engagement in online electronic word-of-mouth communication. This, in turn, helps virtual influencers generate more traffic.

This study also has important practical implications for virtual influencers and their associated brands

- The AI virtual influencers like Liu Yexi and related brands should enhance promotion of influencers. The results show Liu had very good social relation and social image, and that advantages need to be maintained. Comparatively, the self-creation and UGC are a little weak, so virtual influencers and brands should improve the quality of the original content, and they can organize some activities to engage more followers in UGC creation.
- The AI virtual influencers and brands need to clear the way for consumers to search for related information. They can provide more useful information to improve the efficiency of consumers' search, and they can use multiple social media platforms to publish information to enhance the information search convenience.
- They can provide some rewards for consumers who share information on social media. The results of this study suggest that people are more willing to share when they can have some material rewards, and their share can be very beneficial for virtual influencers' popularity. So virtual influencers and brands should make good use of this great way to increase online electronic word-of-mouth.

Though this study makes aforementioned contributions, there are still some limitations. The sampling range can be enlarged. This study included 306 Chinese social media users who were familiar about Liu Yexi, and more valid data from social media users all around the world can make more convincing conclusions. Besides, this study based on the AISAS theory to investigate consumers' responses and virtual influencers' marketing effectiveness. However, there are some other indicators which were not mentioned in this paper can be useful.

6 CONCLUSION

The purpose of this paper is to clarify the current situation of virtual influencers' promotion and consumer reactions to virtual influencer marketing, in order to provide theoretical insights and practical suggestions for optimizing their marketing strategies. Based on the AISAS model, this study structured a reseach framework to include the promotion of virtual influencer, consumers' attention, interest, search willingness, purchasing action, and desire to share. And by using SPSS and AMOS to analyze, we got some important findings which verified the effectiveness of AI virtual influencer on consumers' attention, interest, and search willingness, and reveal the most important factor which can have positive influence on consumers' purchasing action is their search willingness. The paper also shows consumers' purchasing actions increase their desire to share which is a kind of electronic word-of-mouth. Additionally, the results also reveal some aspects of virtual influencers' promotion can be improved, which will be helpful to promote virtual influencers' marketing effectiveness.

This paper contributes to the understanding of virtual influencers' role in social media marketing, highlighting their potential to influence consumer decisions and the importance of optimizing promotional strategies to enhance marketing effectiveness. The study underscores the need for brands to focus on comprehensive and engaging content to foster higher consumer interaction and conversion rates.

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